

Applying Exponent LawsQuestion Bank - Additional 3.05 **ANSWER KEY**

*Show your work. Circle your final answer.***For questions 1 through 4, rewrite each expression as an equivalent expression with a positive exponent.**

1. $\left(-\frac{4}{9}\right)^{-5}$

$\left(-\frac{9}{4}\right)^5$

2. $\left(\frac{9}{10}\right)^{-6}$

$\left(\frac{10}{9}\right)^6$

3. $\left(\frac{-1}{2}\right)^{-8}$

$(-2)^8$

The reciprocal of $-1/2$ is -2 .

4. $\left(\frac{2}{7}\right)^{-3}$

$\left(\frac{7}{2}\right)^3$

Matching.

5. Match each expression on the left with the equivalent expression on the right by writing the letter.

a. $\left(-\frac{13}{9}\right)^{-3}$ A. $\left(\frac{13}{9}\right)^3$ B. $\left(-\frac{13}{9}\right)^3$ C. $\left(-\frac{9}{13}\right)^3$ D. $\left(\frac{9}{13}\right)^3$

C

b. $\left(\frac{13}{9}\right)^{-3}$

D

c. $\left(-\frac{9}{13}\right)^{-3}$

B

d. $\left(\frac{9}{13}\right)^{-3}$

A

For questions 6 through 9, generate an equivalent expression with one base and one positive exponent. Show the laws you use.

6. $\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$
 $\left(-\frac{5}{4}\right)^{-10} = \left(-\frac{4}{5}\right)^{10}$

7. $\left(\frac{3}{7}\right)^{-5} \div \left(\frac{3}{7}\right)^{-2}$
 $\left(\frac{3}{7}\right)^{-3} = \left(\frac{7}{3}\right)^3$
 $-5 - (-2) = -3$, then flip.

8. $\left[\left(\frac{4}{5}\right)^{-2}\right]^3 \cdot \left(\frac{4}{5}\right)^8$
 $\left(\frac{4}{5}\right)^{-6} \cdot \left(\frac{4}{5}\right)^8 = \left(\frac{4}{5}\right)^2$

9. $\left(\frac{1.5^4}{1.5^2}\right)^{-1}$
 $1.5^{-2} = \left(\frac{3}{2}\right)^{-2} = \left(\frac{2}{3}\right)^2$

Equivalent or not.

10. Consider the expression $\frac{\left(\frac{1}{4}\right)^3}{4^{-5}} \cdot 4^4$. For each expression below, decide whether it is equivalent, and justify with a law of exponents.

a. $\frac{4^5 \cdot 4^4}{4^3}$
Yes — both simplify to 4^6 .

b. $4^{-8} \cdot 4^4$
No — this is 4^{-4} .

c. $\left(\frac{1}{4}\right)^{-2} \cdot 4^4$
Yes — $4^2 \cdot 4^4 = 4^6$.

d. 4^6
Yes.

11. Which is the value of $\left(-\frac{3}{10}\right)^{-3}$?

☐ A. $\frac{27}{1000}$

☐ B. $\frac{1000}{27}$

☐ C. $-\frac{27}{1000}$

☒ D. $-\frac{1000}{27}$

D

12. Find the value of n that makes each statement true.

a. $\left(\frac{2}{7}\right)^n = \frac{49}{4}$

$n = -2$

b. $\left(\frac{1}{3}\right)^n = 27$

$n = -3$